

Financial Economics

Bond Project

Instructions

Complete the following steps of the project. In preparation to how you will be evaluated, show your work clearly, and paste the relevant R (or other software you feel most comfortable with) code parts or functions used, output and AI prompts (if any used) directly into your write-up. Your write-up can be a document or a presentation. We will not grade the write-up directly; you will discuss your findings in an individual meeting with the instructor or TA.

You can imagine the meeting as having to explain your work to a higher-up to ensure your work is in place before a whole presentation presenting all your findings. You will have access to your work so make sure everything is saved, accessible and organized for the meeting.

You are encouraged to (should) use AI tools to assist with the coding. You must be able to explain (in broad terms) what the code does, checks you have performed to ensure its correctness, and what the results mean. You must include code checks by using functions like those under the `assertthat` package in R. Use them to verify assumptions about your data and calculations. As much as possible, visualize your data and output to better understand what the code is doing and to catch potential errors.

In class, I'll use VS Code with Copilot to assist with coding in R and debugging. You are encouraged to do the same or use any other tools you are comfortable with, as long as you can clearly explain your code and its output. Instructions to set up the environment I will use are [here](#). Example code and data to produce the output using bonds selected by the instructor is provided [here](#) in the course GitHub repo. You can adapt this code (with AI assistance) to your own bond selection and analysis.

Question 1: Bond Price Graph

- a. Follow the directions in the course GitHub repo under `guides/FactSet_Access_and_Company_Selection.md` ([link](#)). You should be able to download bond price and yield to maturity data for five companies from FactSet Web. Remember the data should be available for a year prior to the end of the month of when the semester starts. For example, if the semester starts in September 2026, you should be able to find bond data for each company from the end of August 2025 through the end of August 2026.

Once you've downloaded the data, first, construct a table with the following columns: Date and the bond price for each of the five companies. The table should be sorted by date. Download the data with as many decimals as possible. Your table should look like the example below:

Date	Company 1 Bond Price	Company 2 Bond Price	...	Company 5 Bond Price
MM/DD/YYYY	X1	Y1	...	V1
MM/DD/YYYY	X2	Y2	...	V2
		...		
MM/DD/YYYY	X[1y data]	Y[1y data]	...	V[1y data]

You can also see a data table sample using the instructor's financial asset choices in the course GitHub repo under `bond-assignment/data/BondPrices.xlsx`.

- b. The data from the five bonds you picked are part of 10 publicly-traded companies that you are tracking this semester. Briefly explain (2–3 sentences) why you chose this particular set of ten companies. What's the thematic connection? There does not need to be one, but if there is a reason for your selection, please share it. Remember it can be any factor such as you like the company products, the company names, company logo colors are similar to Columbia's, etc.
- c. Bond price data from FactSet Web follows the market convention that prices are quoted as a percentage of face value. For example, a price of 98.50 means \$985 per \$1,000 face value or \$4925 per \$5,000 face value. Write out an example of how to convert a bond price from the FactSet percentage (market convention) to the actual dollar price for a bond you picked on a particular date.

Notes: Face value is typically \$1,000 per bond for U.S. corporate bonds, but verify this in FactSet for the bond you picked. Have screenshots of the bond information in case you are asked about them in the meeting.

- d. Create a line graph of the bond prices (as given by FactSet when you downloaded the data, as percentage of face value) for the five companies over the one-year period. The x-axis should represent the date, and the y-axis should represent the bond price in dollars. Each company's bond price should be represented by lines that are easy to distinguish and include a legend to identify each company.

Question 2: Bond Yield Graphs

- a. Similar to Question 1, now download the bond yield to maturity data for the five bonds from different companies from FactSet Web. You should have a table with the following columns: Date and the bond yield to maturity for each of the five companies. Download the data with as many decimals as possible. The table should be sorted by date.

You can also see a data table sample using the instructor's financial asset choices in the course GitHub repo under `bond-assignment/data/BondYields.xlsx`.

- b. Create a line graph of the bond yields for the five companies over the one-year period. The x-axis should represent the date, and the y-axis should represent the bond yield in percentage points. Each company's bond yield should be represented by lines that are easy to distinguish and include a legend to identify each company.

Question 3: Interpretation of your Graphs

- a. In light of your bond price and yield graphs, explain the relationship between bond prices and yields. What is the economic reasoning behind the relationship between bond prices and yields? Use your graphs to illustrate your answer.
- b. Are there any notable trends in common across the five bonds? For example, do they *all* move in the same direction over a time period? Is there a sudden peak or drop across *all* bonds? Discuss potential reasons for these trends. You may use news articles, press releases, or other publicly available information as supportive evidence.
- c. Focus on the performance for one bond of your choice. Describe any particular change in its price or yield over the year. What might be some reasons for this change? Provide sources to support your reasoning (e.g., news articles, press releases, or other publicly available information). Note that you are not expected to do a full credit analysis, but rather to provide a plausible explanation for the observed trend with some evidence.

Question 4: Yield to Maturity-Price Calculation

Let's confirm we feel comfortable with the bond YTM and price relationship. For one of the bonds from Question 2, pick a specific date that's exactly an integer number of semi-annual periods (e.g., 6 months, 12 months, 18 months, etc.) from the bond's maturity date if the bond makes semi-annual coupon payments.

Apply the bond pricing equation to solve for the bond price for the selected date supposing you didn't know the price and only knew the yield. You know the bond's coupon rate, face value, yield to maturity, and number of quarters to maturity. For example, if a bond has a 6% coupon rate, face value \$1,000, current YTM 7%, and 1 year to maturity, you would write to solve for the price as follows:

$$\$990.50 = \frac{30}{(1 + \frac{YTM}{2})} + \frac{30 + 1000}{(1 + \frac{YTM}{2})^2}$$

The bond price you obtain from this calculation should be almost identical to the bond price you retrieved from FactSet Web for that date (after the adjustment of bond market prices quoted as percentage value of face value).¹ Show your work, take screenshots of the bond information you used for the calculation, and any code or Excel/calculator functions and inputs you used to verify your calculation.

¹. It may differ slightly due to rounding or other minor methodological differences.

Question 5: Comparative Table and Chart

Part A: Summary Table

Compile a single summary table with all bonds from Question 2, sorted from lowest to highest YTM as of the last date available in the data you downloaded. The table should include: Company, Coupon Rate (%), Maturity Date, Years to Maturity (as of last date in the data), Market Price (\$), and YTM (%). Remember that a company will have many bonds, but you are working with one bond per company for five companies.

Part B: Chart

Create a bar chart or dot plot displaying the YTM of each bond on the vertical axis and the company name on the horizontal axis, ordered from lowest to highest YTM. Label each bar/point with the YTM value.

Include the chart as a figure in your write up. Make sure the axes are clearly labeled and the chart has a descriptive title.

Part C: Interpretation

- a. From your sorted table in Question 5, Part A, identify the two bonds whose maturity dates are closest to each other. Report the company names, maturity dates, and YTMs of the two bonds.
- b. Write a few sentences interpreting the yield differences or similarities between these two bonds. Among factors to consider:
 - Companies' credit risk and bond ratings
 - Industry or sector
 - Bond coupon rates
 - Degree to which maturity affects the yield

Deliverables For Meeting

You will be graded based on an individual meeting. You will submit your report ahead of the meeting so the instructor can review it before the meeting. Beyond the answers to questions above, please have ready the following items ready as potential discussion points. You may be asked to share your screen and walk through (parts of) your code, figures, and tables.

1. **Format:** A single PDF report (document or presentation) compiling your output and main points answering the project's questions. No page limit, but do not paste thousands of rows of code or text. As an appendix, include screenshots or text excerpts of the AI prompts you found most useful when working on the project. You will upload the PDF report in Courseworks by 11/1 before the meeting.
2. **Figures:** Embed all figures with numbers, captions and axis labeled.
3. **Tables:** Embed all tables with numbers and captions.
4. **Code:** Have your code available to share. You may be asked to walk through parts of it.
5. **AI Usage:** Save your prompts and the AI's responses in an appendix (via saving the conversation as a text file or through screenshots). You may be asked to discuss how you used AI, what you learned from it, and how you verified its output.